

Lab 2 — Shaping Data & Advanced Visuals in Power BI

Append • Hierarchies • Multi-row Card • Sorting • Legends • Secondary Axis • Filters in Depth

1. Lab Overview & Learning Outcomes

In Lab 1 you loaded data, built relationships, and made cards, tables and slicers. In Lab 2 you go further: you will **combine (append) three files into one table**, build a **hierarchy** you can drill into, and create richer visuals using a **multi-row card**, **sorting**, **legends** and a **secondary y-axis**. Finally you will take a deep dive into **filters** — the single most important skill for controlling what a dashboard shows.

By the end of this lab you will be able to:

- Append several files with the same structure into one combined table.
- Create a hierarchy and drill up/down inside a visual.
- Use a multi-row card to display several values at once.
- Sort a visual by value, and fix wrong label order with Sort-by-column.
- Split a visual into series using a Legend.
- Show two different scales on one chart using a secondary y-axis.
- Apply visual-level, page-level and report-level filters, plus Top N and Advanced filtering.
- Apply your own formatting to make a dashboard clean and professional.

2. The Dataset — U.S. Business Dynamics

All files describe U.S. manufacturing businesses: how many firms and establishments exist, how many employees they have, and how jobs are created and destroyed over time. You will use them to practise data shaping and visuals.

File	What it contains
Establishment_Survey.csv	Main table. Firms, establishments and employees by industry and year (1978–2018). Includes an Industry hierarchy: Sector → Subsector → Industry group → NAICS.
EstablishmentAge.csv	Small lookup table: Age code → Establishment Age label (e.g. 0 years, 6-10 years, 26+ years). Contains duplicate rows you will remove.
manufacturing_data.csv	Sales, payroll and employees by U.S. state and industry (2018–2019). Has an extra description row under the header to remove.
Timeseries_1979.csv	Three files, one per year, identical columns . Job creation/destruction for U.S. manufacturing. These are the files you will APPEND.
Timeseries_1980.csv / _1981.csv	Same structure as 1979 — the other two files to append.
Time.txt	A time dimension: Year, Decade, Century (1961–2021). Tab-separated. Used for a Century → Decade → Year hierarchy and for relationships.
NEW CONCEPT — APPEND vs MERGE: Append stacks rows on top of each other (same columns, more rows) — perfect for yearly files. Merge joins columns side-by-side using a key (like a relationship). Today you will Append.	

Part A — Load the Files & Quick Clean (≈ 7 min)

1. Open a new Power BI Desktop file. Using **Get data** → **Text/CSV**, load: `Establishment_Survey.csv`, `EstablishmentAge.csv`, `manufacturing_data.csv`, `Time.txt`, and all three `Timeseries_1979/1980/1981.csv` files.
2. For `Time.txt`, Power BI will detect a **Tab** delimiter automatically — just click **Load**.
3. Open **Transform data** to enter Power Query. Select the **manufacturing_data** query: its first data row is a description (“id, Geographic Area Name...”). Remove it with **Home** → **Remove Rows** → **Remove Top Rows** → **1**.
4. Select the **EstablishmentAge** query — it has repeated rows. Click **Home** → **Remove Rows** → **Remove Duplicates**. You should be left with 13 unique age codes.
5. In **EstablishmentAge**, click the type icon on the Age code column and set it to **Whole Number** (you will need this for sorting later).

Part B — Append the Three Yearly Files (≈ 10 min)

Each Timeseries file also has a description row under its header. Clean all three first, then stack them.

B1. Clean each yearly file

1. Select the **Timeseries_1979** query. Use **Home** → **Remove Rows** → **Remove Top Rows** → **1** to delete its description row.
2. Repeat the exact same step for **Timeseries_1980** and **Timeseries_1981**.

B2. Append into one table

3. On the **Home** ribbon click **Append Queries** ▾ → **Append Queries as New** (“as New” keeps your originals and creates a combined copy).
4. Choose **Three or more tables**. Add `Timeseries_1979`, `1980`, `1981` to the right-hand list, then **OK**.
5. Rename the new query (double-click it in the Queries list) to `JobDynamics`. It should now have all three years stacked together (about 39 rows).

B3. Set data types on JobDynamics

6. Set `YEAR` to **Whole Number**.
7. Set the count columns (`FIRM`, `ESTAB`, `EMP`, `ESTABS_ENTRY`, `ESTABS_EXIT`, `JOB_CREATION`, `JOB_DESTRUCTION`, `NET_JOB_CREATION`) to **Whole Number**.
8. Set the rate columns (anything ending in `_RATE`) to **Decimal Number** — you will need these for the secondary axis.
9. Click **Home** → **Close & Apply**. All tables, including `JobDynamics`, load into the report.

CHECKPOINT ✓ In the Data pane you should now see JobDynamics with a YEAR column containing 1979, 1980 and 1981. If a year is missing, re-open Power Query and check that file was added to the append.

Part C — Relationships & a Drill-down Hierarchy (≈ 9 min)

C1. Relationships

Switch to **Model view** and create these relationships by dragging the field from the first table onto the matching field in the second:

From (dimension)	To	Type
Time[Year]	Establishment_Survey[Year]	1 → *
Time[Year]	manufacturing_data[YEAR]	1 → *
Time[Year]	JobDynamics[YEAR]	1 → *
EstablishmentAge[Age code]	Establishment_Survey[Age code]	1 → *

The **Time** table is a shared date dimension — one small table filtering three larger ones. That is exactly how a data warehouse reuses dimensions.

C2. Create a hierarchy

A **hierarchy** lets one field in a visual drill into more detail. Build a calendar hierarchy on the Time table.

1. In the **Data pane**, expand **Time**. Right-click the Century field → **Create hierarchy**.
2. A new **Century Hierarchy** appears. Rename it Calendar (right-click → Rename).
3. Drag Decade onto the Calendar hierarchy, then drag Year onto it too. Order should read Century ► Decade ► Year.

C3. Use it in a chart with drill-down

4. On the report page add a **Clustered column chart**. Drag the **Calendar** hierarchy to **X-axis** and Establishment_Survey[Number of establishments] to **Y-axis**.
5. Add a **visual-level filter** (Filters pane): set **Establishment Age** (from EstablishmentAge) to **Total** so you count each establishment once. (More on filters in Part G.)
6. Use the arrows at the top of the visual to drill: the **double-down arrow** goes to the next level; the **up arrow** goes back. Drill Century → Decade → Year and watch the bars split.

WHY THIS MATTERS: Hierarchies turn one chart into many. Instead of building separate visuals for year, decade and century, users drill the same chart to the detail they want.

Part D — Multi-row Card & Sorting (≈ 8 min)

D1. Multi-row card

A **multi-row card** shows several values as a list — great for a compact breakdown.

1. Click a blank area → **Multi-row card** visual (in the Visualizations pane).
2. Drag in Establishment_Survey[Subsector] then Establishment_Survey[Number of establishments]. You now see each subsector with its own value.
3. Tip: you can also drop several measures (e.g. Number of firms, establishments, employees) with no category to show a KPI stack.

D2. Sort a visual by value

1. Select the multi-row card (or make a bar chart of the same fields). Click the visual's ... **(More options)** → **Sort axis / Sort by** → choose **Number of establishments**, then set **Sort descending**.

D3. Sort-by-column (fix wrong label order)

Put **Establishment Age** on a bar chart and you will see it sorts oddly (“0 years, 1 year, 11-15 years, 16-20...”) because text sorts alphabetically. Fix it so it follows the real age order.

2. Go to **Table view** (or Data view). In the Data pane click the **Establishment Age** column.
3. On the **Column tools** ribbon click **Sort by column** → choose **Age code**.
4. Return to your report — **Establishment Age** now orders by its code (Total, 0 years, 1 year, 2 years ... 26+ years, Left Censored) in every visual and slicer.

REMEMBER: Sort-by-column needs a numeric column that has one value per label. That is why you set Age code to Whole Number back in Part A.

Part E — Legends (≈ 6 min)

A **Legend** splits one visual into colour-coded series so you can compare groups.

1. Add a **Clustered column chart** using **manufacturing_data**: X-axis = NAME (state), Y-axis = Number of employees (EMP).
2. Add a **visual filter**: **Meaning of NAICS code (NAICS2017_LABEL) is Manufacturing**, so each state is counted once at the top level (not summed across every industry).
3. Drag YEAR into the **Legend** well. Each state now shows two bars — 2018 and 2019 — in different colours.
4. Open the **Format** pane → **Legend** to move it (top / right) or rename its title.

LEGEND vs AXIS: The X-axis says what the bars are (states); the Legend says how each bar is split (by year). Swap a field between them to see the difference.

Part F — Secondary Y-axis (Combo Chart) (≈ 8 min)

When two measures use very different scales — a big **count** and a small **percentage** — put one on a **secondary y-axis** so both are readable. Your appended JobDynamics table is perfect for this.

1. Add a **Line and clustered column chart** (Visualizations pane).
2. Set X-axis = YEAR (the appended years 1979–1981).
3. Add a **visual filter**: **EAGE_LABEL is Total**, so you get one value per year.
4. Set Column y-axis = JOB_CREATION (large counts).
5. Set Line y-axis = JOB_CREATION_RATE (small %). Power BI automatically adds a **second axis on the right** for the line.
6. Open **Format** → **Y-axis** and check the **Secondary** section — you can rename it, change its number format, or hide it.

READING IT: Bars (left axis) show how many jobs were created each year; the line (right axis) shows the rate. Without the second axis the line would look flat against the millions on the left.

Part G — Filters in Depth (≈ 12 min)

The **Filters pane** (right side of the screen) is the most powerful way to control a report. It has **three scopes** and several **filter types**. Learn these well — you will use them constantly.

The three filter scopes

Scope	What it affects
Filters on this visual	Only the one selected visual. Use for tweaks like “this chart shows Total only.”
Filters on this page	Every visual on the current page. Use for “this page is about 2018.”
Filters on all pages	Every visual on every page of the report. Use for “the whole report is Manufacturing.”

The filter types

Type	What it does
Basic	Tick/untick values from a list (e.g. Year = 2018).
Advanced	Conditions such as is greater than, contains, starts with (e.g. Number of employees > 1,000,000).
Top N	Keep only the top or bottom N items by a chosen value (e.g. Top 5 Subsectors by establishments).
Relative	For dates/numbers relative to now (e.g. in the last 3 years). Available on date fields.

G1. A visual-level filter

1. Select your column chart. In the Filters pane, under **Filters on this visual**, find **Year** and set it to **2018** only. Only that visual changes.

G2. A page-level filter

1. Click an empty part of the page. Drag Establishment_Survey[Sector] into **Filters on this page** and tick **Manufacturing**. All visuals on the page obey it.

G3. A report-level (all pages) filter

1. Drag a field such as Time[Century] into **Filters on all pages** to constrain the entire report. (Add a second page to prove it carries over.)

G4. Top N filtering

1. Make a bar chart: Y-axis = Subsector, X-axis = Number of establishments.
2. In its visual filter for **Subsector**, change **Filter type** to **Top N**. Set Show items = Top 5, and drag Number of establishments into **By value**. Click **Apply filter**.

G5. Advanced filtering

1. On any visual with **Number of employees**, set its filter type to **Advanced**, choose **is greater than**, type a threshold (e.g. 1000000), and **Apply filter**.

Housekeeping

- **Clear a filter** with the eraser icon on its filter card; **expand/collapse** cards with the arrow.
- **Hide a filter card** (eye icon) so viewers can't change it, or **lock** it, in the Format pane for the Filters pane.
- **Slicer vs filter**: a slicer sits on the canvas for the audience to click; a filter lives in the Filters pane and is usually set by you, the author.

PRACTICE ✓ Combine scopes: keep the report on Manufacturing (all pages), the page on 2018, and one visual on Top 5 subsectors. Notice how they stack — each scope narrows the data further.

Part H — Make It Beautiful (On Your Own)

Use your remaining time to turn your visuals into a dashboard you would be proud to present. Work independently, there is no single right answer, so make your own design choices.

Key Terms (Glossary)

Append	Stacking rows from tables with the same columns into one longer table.
Merge	Joining columns from two tables using a matching key (like a relationship).
Hierarchy	An ordered set of fields (e.g. Century ▶ Decade ▶ Year) you can drill up and down.
Drill down / up	Moving a visual between hierarchy levels to show more or less detail.
Multi-row card	A visual that lists several values or a category breakdown as rows.
Sort-by-column	Ordering a text column by the values of another (usually numeric) column.
Legend	A field that splits a visual into colour-coded series.
Secondary y-axis	A second vertical scale (on the right) for a measure with a different range.
Filter scope	Whether a filter affects one visual, one page, or all pages.
Top N filter	Keeps only the highest (or lowest) N items by a chosen value.

Appendix — Troubleshooting

Appended table is missing a year	Re-open Power Query → click the JobDynamics (Append) step → confirm all three queries are listed; re-add the missing one.
Description text appears as a data row	You forgot Remove Top Rows → 1 on that file. Do it, then the append updates automatically.
Sort-by-column is greyed out / errors	Age code must be a number with one value per label. Set it to Whole Number and remove duplicates first.
Combo chart line looks flat	The rate is tiny next to the count — make sure the rate is on the Line (secondary) axis, not the column axis.

Numbers look far too big	You are summing across totals and detail rows. Add a visual filter (e.g. Age = Total, or NAICS = Manufacturing).
A filter isn't doing anything	Check its scope — a visual filter only affects the selected visual; click Apply filter if using Advanced/Top N.

End of Lab 2 — great work! You can now shape data and build advanced, filtered visuals.